

EBN122 EKSAMEN/EXAMINATION (NOVEMBER 2016)

ANTWOORDBLAD / ANSWER SHEET

Surname / Van:	Memo.	Student #:							
Initials/ Voorletters:									

Q1	18	Q2	32	Q3	20	Q4	24	Tot:	/90
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QUESTION 1 / VRAAG 1 (18)		
1.1	a) $p(35s) = +13.125W$ ✓✓	b) $Q = 40C$ ✓✓
1.2	$I_1 = 0.06A$ ✓✓	$V_2 = -1.2V$ ✓✓
1.3	$R = 16\Omega$ ✓✓	
1.4	$R_{cd} = 155\Omega$ ✓✓	
1.5	$I_S = -0.55A$ ✓✓	$V_1 = 1.875V$ ✓✓

QUESTION 2 / VRAAG 2 (32)				
2.1	$a = 18$ ✓	$b = 6$ ✓	$c = -10$ ✓	$d = 9$ ✓
2.2	$e = -1$ ✓	$f = 1$ ✓	$g = 5$ ✓	$h = 28$ ✓
2.3	$V_Q = -16V$ ✓✓			
2.4	a) $I_B = 0.1mA$ ✓✓	b) $V_{CE} = 11.58V$ ✓✓		
2.5	a) $V_{O(1A)} = -20V$ ✓✓	b) $V_{O(30V)} = 60V$ ✓✓		
2.6	a) $R_T = 6k\Omega$ ✓✓	b) $V_X = 10V$ ✓✓		
2.7	a) $V_{TH} = 7.5V$ ✓✓	b) $R_{TH} = 15\Omega$ ✓✓		
2.8	a) $I_N = -1.5A$ ✓✓	b) $R_N = 40\Omega$ ✓✓		
2.9	$V_{TH} = \pm 12V$ ✓✓			

QUESTION 3 / VRAAG 3 (20)

3.1	$i_0 = 5A \checkmark$		
3.2	a) $V_0 = -16V \checkmark$		b) $V_0 = -39V \checkmark$
3.3	a) $V_m = 6V \checkmark$	b) $V_n = 2V \checkmark$	c) $V_o = 6V \checkmark$
3.4	$v_C(t) = 8.25V \checkmark$		
3.5	$C_{eq} = 10F \checkmark$		
3.6	$w_C = 0J \checkmark$		$w_L = 9J \checkmark$

VRAAG 4 – QUESTION 4 (24)

4.1	a) $\bar{V} = 3.54 + j17.68V \checkmark$ (rectangular)	
	b) $v_C(t) = 2\cos(10^6t - 155^\circ)V \checkmark$	
	c) $R = 6.5\Omega$ or $L = \checkmark$ or $C = \checkmark$	
4.2	a) $\bar{I}_1 = 2.5 \angle 8.13^\circ A \checkmark$ (polar)	
	b) $\bar{V}_1 = 0.5 \angle 90^\circ \checkmark$ (polar)	
4.3	$\bar{Z}_{ab} = 40 - j40\Omega \checkmark$ (rectangular)	
4.4	$a_1 + jb_1 = 1 \checkmark$	$a_2 + jb_2 = -1 \checkmark$
	$a_3 + jb_3 = -2 \checkmark$	$a_4 + jb_4 = 2 + j \checkmark$
4.5	$\bar{V}_{Th} = 1.414 \angle 0^\circ V \checkmark$ (polar)	$\bar{Z}_{Th} = 3.54 \angle -45^\circ \Omega \checkmark$ (polar)
4.6	$\bar{I}_N = 7.07 \angle -45^\circ A \checkmark$ (polar)	$\bar{Z}_N = 7.07 \angle -45^\circ \Omega \checkmark$ (polar)